

4.1 Maclaurin & Taylor Polynomials Worksheet

1. Given a function $f(x)$ for which

$$f(2)=3, \quad f'(2)=5, \quad f''(2)=8, \quad f'''(2)=-18, \quad \& \quad f^{(4)}(2)=240,$$

write out the degree 4 Taylor polynomial for $f(x)$ centered at $x=2$.

2. Find the 3rd Taylor polynomial (that is, find p_3) for $f(x)=\sqrt{x}$ about $x=4$.

3. Consider the function $f(x) = \tan^{-1}(x)$.

a) Find the 3rd Maclaurin polynomial for $f(x) = \tan^{-1}(x)$.

b) Use the polynomial you found in part (a) to provide an estimate of the value $\tan^{-1}(0.25)$.

4. What is the coefficient of x^2 in the Maclaurin polynomial (which has degree 2 or higher) for

$$f(x) = \frac{x-2}{6x^2+x-1} ?$$

5. Use a Taylor polynomial of degree 2 to obtain a good estimate for $\ln(1.2)$. You don't need to simplify your answer.

6. Use a Taylor polynomial of degree 2 to approximate the value of $\sqrt{103}$.